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A palaeomagnetic study of Jurassic intrusives from southern New South Wales: further evidence for a pre-Cenozoic dipole low

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SUMMARY

The results of rock magnetic, thermal demagnetization and Thellier palaeointensity studies are presented for two high-level intrusions from southern New South Wales, Australia. The Gingenbullen Dolerite (GB) (34.5°S, 150.3°E) and the Gibraltar Microsyenite (GS) (34.5°S, 150.4°E) were emplaced during the second of three major phases of igneous activity that affected the southern Sydney Basin and are K/Ar dated at 172 and 178 Ma, respectively. The magnetic mineralogy of the two intrusions is different: the GB is dominated by single-domain (SD)/pseudo-single-domain grains of magnetite, whereas the GS has both SD magnetite and haematite components, although the haematite component does not record a stable remanence. The GB records two opposed components of magnetization, above 100 °C, occupying distinct parts of the blocking temperature spectrum. The lower blocking temperature (LBT) component resides between 150 and 400 °C and is of normal polarity, having a mean direction of $D/I = 021°/-79°$ ($\alpha 95 = 8°$, $k = 63$), with a corresponding VGP at Lat/Long = 54°S/137°E ($dp = 12$, $dm = 13$). This component is believed to be a TCRM or TVRM related to the initial opening of the Tasman Sea at ~90 Ma. The higher blocking temperature (HBT) ChRM component is recognized between 450 and 580 °C and is of reverse polarity, with a mean direction of $D/I = 174°/+81°$ ($\alpha 95 = 11°$, $k = 33$) and a corresponding VGP at Lat/Long = 52°S, 153°E ($dp = 17$, $dm = 18$). The ChRM for the GS resides in the 100–450 °C region and, although the scatter is high, has a similar mean direction ($D/I = 167°/+79°$; $\alpha 95 = 24°$, $k = 26$) to the HBT component of the GB, with a pole position at Lat/Long = 54°S/158°E ($dp = 30$, $dm = 31$). There is no evidence of the GB LBT component in samples from the GS, suggesting that this intrusion was magnetically unaffected by the later event, which is also seen for other intrusions in the Sydney Basin. The VGPs calculated from the ChRM directions both plot on the Jurassic segment of the Australian APWP (Dunlop *et al.* 1997), providing further palaeomagnetic confirmation for the K/Ar ages. Thellier palaeointensity results were of variable quality, as expected for rocks of this age. The mean VDM values calculated for the ChRMs of the GB and GS were 1.3×10^{22} and 4.1×10^{22} Am^2, respectively (approximately 16 and 51 per cent of the present value). On application of a cooling-rate correction estimate, the GB result falls to a minimum of 0.87×10^{22} Am^2 (approximately 11 per cent of the present-day value), representing an overestimate of up to 56 per cent. These values, arising from a time window previously lacking in Thellier estimates, provide further confirmation of the existence of a pre-Cenozoic dipole low. The VDM values derived from the overprint in the GB samples yield an average of 4.8×10^{22} Am^2 (from a range between 44 and 81 per cent of the present-day mean VDM) and are tentatively attributed to the Cretaceous Normal Superchron (CNS). These results may document evidence for a previously unrecognized low dipole moment during the bulk of the CNS, a possibility that requires further investigation by performing palaeointensity work on igneous rocks with radiometric ages between 110 and 90 Myr that carry unequivocal primary TRMs.

Key words: Australia, Jurassic, palaeointensity, pre-Cenozoic dipole low.

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1 INTRODUCTION

The global palaeointensity record [Fig. 1, compiled from the global palaeointensity database (Perrin 1998) supplemented by other published work; see Fig. 1 caption] is dominated by a feature known as the Mesozoic Dipole Low (MDL; Bol'shakov & Solodovnikov 1981; Prévot *et al.* 1990) that documents low values of the virtual dipole moment (VDM) (25–80 per cent of the present-day value) for a considerable period during the Phanerozoic. Dating the extremities of this feature has proved difficult due to the paucity of data for the Permo-Triassic and Mid–Late Cretaceous (e.g. Pick & Tauxe 1993; Thomas *et al.* 1998). Recent studies suggest that the MDL may extend back to the late Palaeozoic and be representative of a generally weak pre-Cenozoic field (Thomas *et al.* 1998). Perrin & Shcherbakov (1997) showed, by detailed statistical analyses of the record, that the geomagnetic field was dipolar during the period of low VDM. The data therefore satisfy the fundamental law of palaeomagnetism, proving the first-order reliability of the record. In addition, they confirm that insufficient sampling of a Neogene-type field could not have produced the MDL. These analyses confirm that the MDL is a real feature and increase our confidence in the reliability of the palaeointensity record. Notwithstanding, there are significant gaps in the palaeointensity record (labelled 1–4 in Fig. 1) that preclude a more detailed definition of the pre-Cenozoic Dipole Low (P-CDL). These gaps in the record occur as a result of inadequate temporal data coverage, due to either a lack of accurate age determinations for target rocks or limited exposure of well-dated rocks. Overcoming these problems is a key focus of current palaeointensity research. Gap 1, covering the period 60–20 Ma, has been recently addressed by Juárez *et al.* (1998) and Solodovnikov (1998). Their data, whilst not completely filling the gap, suggest that the low dipole field also lasted for a considerable part of the Cenozoic, providing further evidence for the existence of a generally weak dipole field for much of Phanerozoic time (e.g. Thomas *et al.* 1998).

The period 250–170 Ma, where data coverage is sparse, is crucial in evaluating whether the P-CDL is a single long feature or two lows, separated by a period of higher field. Much of the data for this period, documented in Fig. 1, are derived from palaeointensity studies of Northern Hemisphere rocks. One of the few studies of Southern Hemisphere rocks (Kosterov *et al.* 1997) supports a low dipole field with palaeointensity values corresponding to a mean VDM of $(4.6 \pm 2.0) \times 10^{22}$ Am^2 (approximately 57 per cent of the present-day value) from 10 lavas within the 180 Ma Karoo Volcanic Province in Lesotho. The present study aims to add to the spatial resolution of the palaeointensity record and provide additional Southern Hemisphere data points for the late part of the 250–170 Ma period. Results of a detailed Thellier study of two large, slowly cooled igneous intrusive bodies from southern New South Wales, Australia are reported and discussed.

2 REGIONAL GEOLOGY AND SAMPLING

The rocks of the Sydney Basin crop out along the coast of central eastern New South Wales and continue offshore across the continental shelf and some 230 km inland. The basement is composed of Lower and Middle Palaeozoic igneous and metamorphic rocks, which are overlain by Carboniferous volcanics and sedimentary rocks in the northern part of the basin. Extensive Permo-Triassic sediments unconformably overlie the basement in the south but are conformable with the Carboniferous sequence in the north. Embleton *et al.* (1985) documented the episodic nature of igneous activity affecting

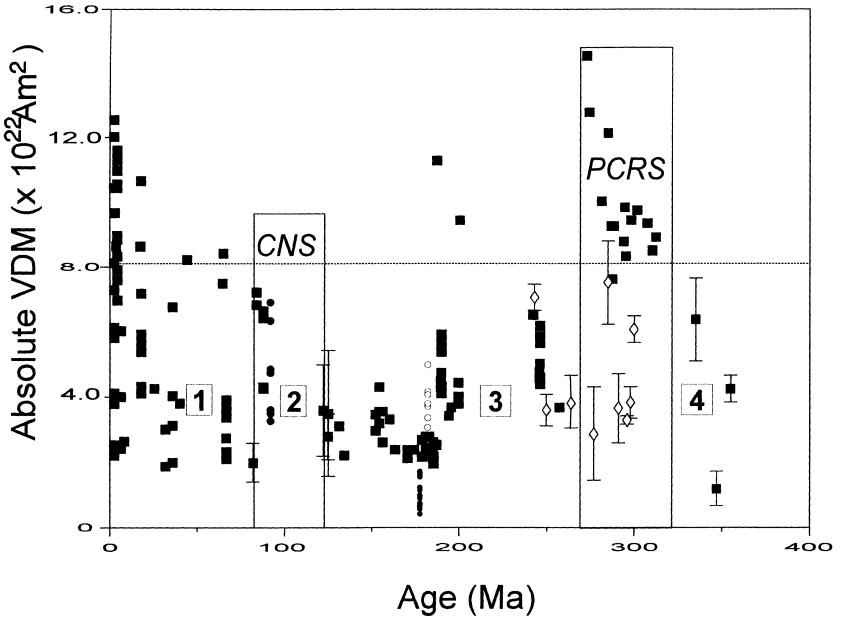


Figure 1. Virtual dipole moment record for the past 400 Myr compiled from the palaeointensity database (Perrin 1998) and Juárez *et al.* (1998) (closed squares) with additional data points from Thomas *et al.* (1998, 1997, 1995); Harcombe-Smee *et al.* (1996) for the Permo-Carboniferous Reverse Superchron (open diamonds) and this study (closed and open circles). Numbered boxes refer to gaps in the palaeointensity record (see text for discussion).

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the Sydney Basin, with three dominant pulses with maxima at around 250, 180 and 50 Ma being responsible for the widespread occurrence of intrusive and extrusive rocks within the basin. The two intrusions sampled in this study were emplaced during the second of these three pulses, which lasted quasi-continuously from 210 to 150 Ma. Other intrusives from the same pulse, located in northern New South Wales, form the subject of a sister study (Thomas & Schmidt, in preparation). Cretaceous activity is largely restricted to the injection of dyke swarms in the Lake Macquarie and Newcastle regions north of Sydney and is probably related to the initial rifting of the Tasman Sea at approximately 90 Ma. Widespread Tertiary basalts and intrusives are found across much of the south-western part of the basin. These are related to the third phase of igneous activity identified by Embleton *et al.* (1985), which probably records the beginning of active separation between Australia and Antarctica (Cande & Mutter 1982).

The Gingenbullen Dolerite (GB) and Gibraltar Microsyenite (GS) intrude the southern part of the Sydney Basin in the Mittagong–Moss Vale region of New South Wales (Fig. 2). The GB is an 80-m-thick horizontal sill, approximately 2 km in length and 1 km wide, composed of crudely columnar quartz dolerite. It intrudes the Wianamatta shales and sandstones at the top of the Liverpool Supergroup and is therefore stratigraphically constrained to be post-Triassic (Boesen *et al.* 1961). This is confirmed by K/Ar dating, which gives a mid-Jurassic age of 172 Ma for the GB (McDougall & Wellman 1976). The sill was sampled at Gingenbullen Mountain quarry, using standard palaeomagnetic drilling equipment, with 43 oriented cores taken from seven sites throughout most of its thickness (see Table 1 for details).

The Gibraltar aegerine-augite microsyenite is a 200-m-thick asymmetric laccolith cropping out on Mt Gibraltar, some 13 km to the northeast of Gingenbullen Mountain. The GS is deuterically altered in places and is characterized by the occasional occurrence of thin (<0.5 m) pegmatite veins. Suitable outcrop on the mountain is now limited due to residential development and intense weathering in some of the disused quarries. However, the intrusion was sampled in four favourable localities throughout its thickness, with 24 oriented cores being collected. The cores were, on average, 10 cm long to enable palaeointensity samples to be taken from unweathered material. The GS intrudes the Permo-Triassic Hawkesbury sandstone and is therefore stratigraphically older than the GB, an observation confirmed by K/Ar dating, which places its age at 178 Ma.

3 ROCK MAGNETIC RESULTS

Rock magnetic experiments were carried out on samples from both the Gingenbullen Dolerite and the Gibraltar Microsyenite in an attempt to establish the magnetic mineralogy of both intrusions and to assess their potential for palaeointensity work.

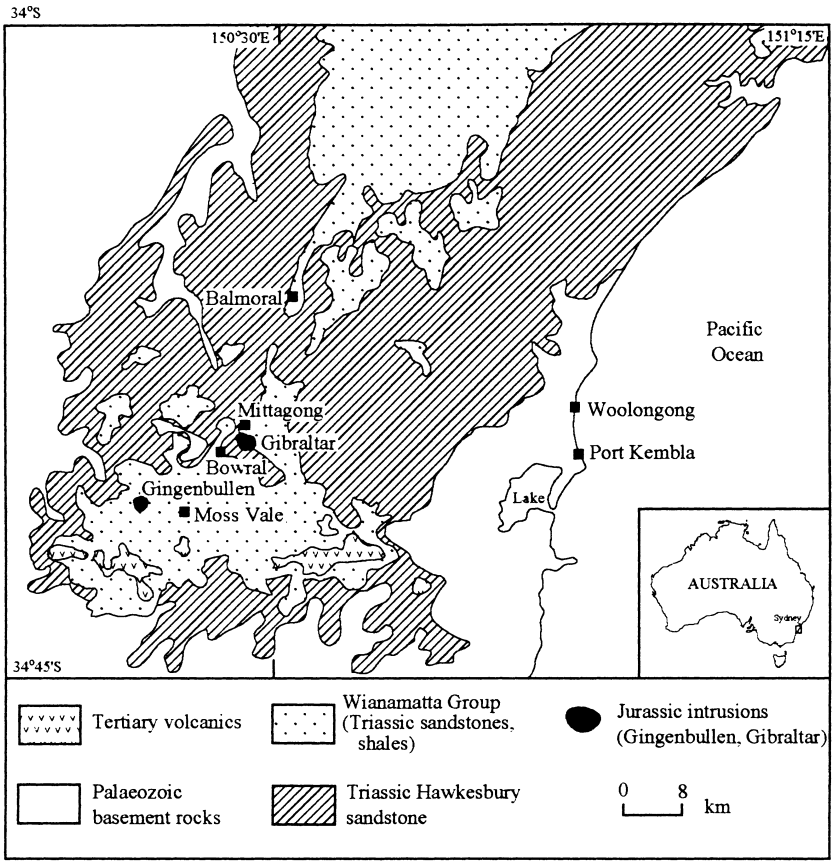


Figure 2. Generalized geological map of the southern Sydney Basin, illustrating the location of the two intrusions sampled in this study (redrawn from the AGSO 1:500 000 sheet of New South Wales).

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Table 1. Results of the thermomagnetic criterion (pTRM tail) rock magnetic test. $pTRM_{(300,\,Tr)}$ and $pTRM_{(500,\,Tr)}$ are the pTRM intensities acquired between 300 °C and room temperature and 500 °C and room temperature, respectively; pTRM $tail_{(300)}$ and pTRM $tail_{(500)}$ are the intensities of the pTRM tail remaining after demagnetization in zero field to 300 and 500 °C, respectively; Tail % indicates the pTRM tail remaining after demagnetization in zero field as a percentage of the original pTRM.

Sample	$pTRM_{(300,\,Tr)}$ ($\times 10^{-6}$ Am^2 kg^{-1})	pTRM $tail_{(300)}$ ($\times 10^{-6}$ Am^2 kg^{-1})	Tail %	Sample	$pTRM_{(500,\,Tr)}$ ($\times 10^{-6}$ Am^2 kg^{-1})	pTRM $tail_{(500)}$ ($\times 10^{-6}$ Am^2 kg^{-1})	Tail %
GB0107B	90.84	6.23	6.86	GB0107B	311.67	3.82	1.23
GB0204B	36.77	3.99	10.85	GB0204B	219.05	4.79	2.19
GB0604B	20.22	3.26	16.12	GB0503B	249.24	7.66	3.07
Average			**11.28**	GB0604B	112.21	2.45	2.18
				GB0705B	51.42	0.84	1.63
				Average			**2.06**
GS0301B	126.40	2.06	1.63				
GS0304B	58.28	1.48	2.54				
GS0403A	143.79	4.96	3.45				
Average			**2.54**				

3.1 Thermomagnetic analyses

Thermomagnetic measurements were performed on small (~0.5 g) samples from each palaeomagnetic site using a translation force Curie balance. Each sample was subjected to a 1.0 T field whilst being heated in air to above its Curie temperature and subsequently cooled back to room temperature. Saturation magnetization was measured continuously [$J_s(T)$] and plotted for both the heating and cooling phases.

The Curie temperature for each sample from the GB was between 570 and 580 °C, indicating pure magnetite as the dominant ferrimagnetic mineral, typically resulting from deuteric oxidation of primary titanomagnetites. Samples from sites GB1–6 showed minimal signs of heating-induced alteration (< 10 per cent rise in J_s), with the cooling curve crossing over the heating curve (Fig. 3a). Site GB7 (Fig. 3b) displayed a lower J_s throughout cooling, suggesting conversion of magnetite to haematite above 500 °C, although again the decrease in J_s was ~10 per cent.

Analyses of the GS revealed Curie temperatures around 650 °C. However, high magnetization values also suggest the presence of magnetite. Site GS1 showed alteration on heating, producing a less magnetic derivative on the cooling phase, probably resulting from oxidation of magnetite to haematite at high temperatures. Site GS2 illustrated similar behaviour to that witnessed in the GB but with a higher Curie temperature, suggesting the possible presence of cation-deficient (CD) magnetite. Site GS3 shows some evidence of an inflection in the heating curve between 300 and 350 °C (Fig. 3e), suggesting the presence of CD magnetite (e.g. Thomas 1993; Orlicky 1990). Samples for $J_s(T)$ analyses were taken from core tops, which show more evidence of alteration. This part of the core was avoided during palaeointensity work, with samples from the bottom sections of cores used to maximize the chances of obtaining a palaeointensity estimate from unweathered material. No evidence of CD magnetite was observed in samples used for palaeointensity analyses of the GS.

3.2 Low-temperature susceptibility (χ_{LT})

χ_{LT} analyses were performed using a Bartington MS2 susceptibility meter in a shielded room. Each sample was cooled to liquid nitrogen temperature (−196 °C) and its susceptibility continuously recorded on warming back to room temperature over a 20 min period.

For samples from the GB, no large changes in susceptibility were witnessed over the specified temperature range, although several samples showed a gradual increase in χ with rising temperature, indicating the presence of single-domain (SD) grains of magnetite (Fig. 4a). No samples from this intrusion showed a peak in χ at the isotropic point of magnetite (~ −150 °C), suggesting that multidomain (MD) grains are not present (Senanayake & McElhinny 1981).

The peak in χ at the isotropic point of magnetite was evident in some of the samples from the GS. However, RS values (χ_{-196}/χ_{RT}) were above 0.5 (e.g. Fig. 4b), suggesting the significant presence of SD magnetite grains (Thomas 1993) in a mixed grain size.

3.3 Hysteresis measurements

The hysteresis properties of samples from both intrusions were measured using a Molspin Vibrating Sample Magnetometer (VSM). Samples from the GB tended to display thin hysteresis loops, with corresponding low coercive force (H_c) and saturation remanence (M_{rs}) values. The ratio of remanent to saturation magnetization (M_{rs}/M_s) is a good indicator of magnetic mineral grain size, since M_{rs} varies according to domain state but M_s is independent of this property. The average value of M_{rs}/M_s for the GB was 0.1, and the GS exhibits broader hysteresis loops (Fig. 5b) with higher M_{rs}/M_s ratios (mean value of 0.2). Fig. 5(c) illustrates a Day plot (Day *et al.* 1977) containing samples from both intrusions. It can be seen that the GB and the GS plot in the PSD region, a feature witnessed in many other studies of ancient igneous rocks.

3.4 IRM acquisition and DC demagnetization

Samples from all four GS sites and three GB sites were exposed to a series of increasing DC fields using a pulse magnetizer until a saturation IRM was obtained. A series of fields were then applied in the opposite sense until the SIRM was reduced to zero. Results are presented as partial remanent hysteresis

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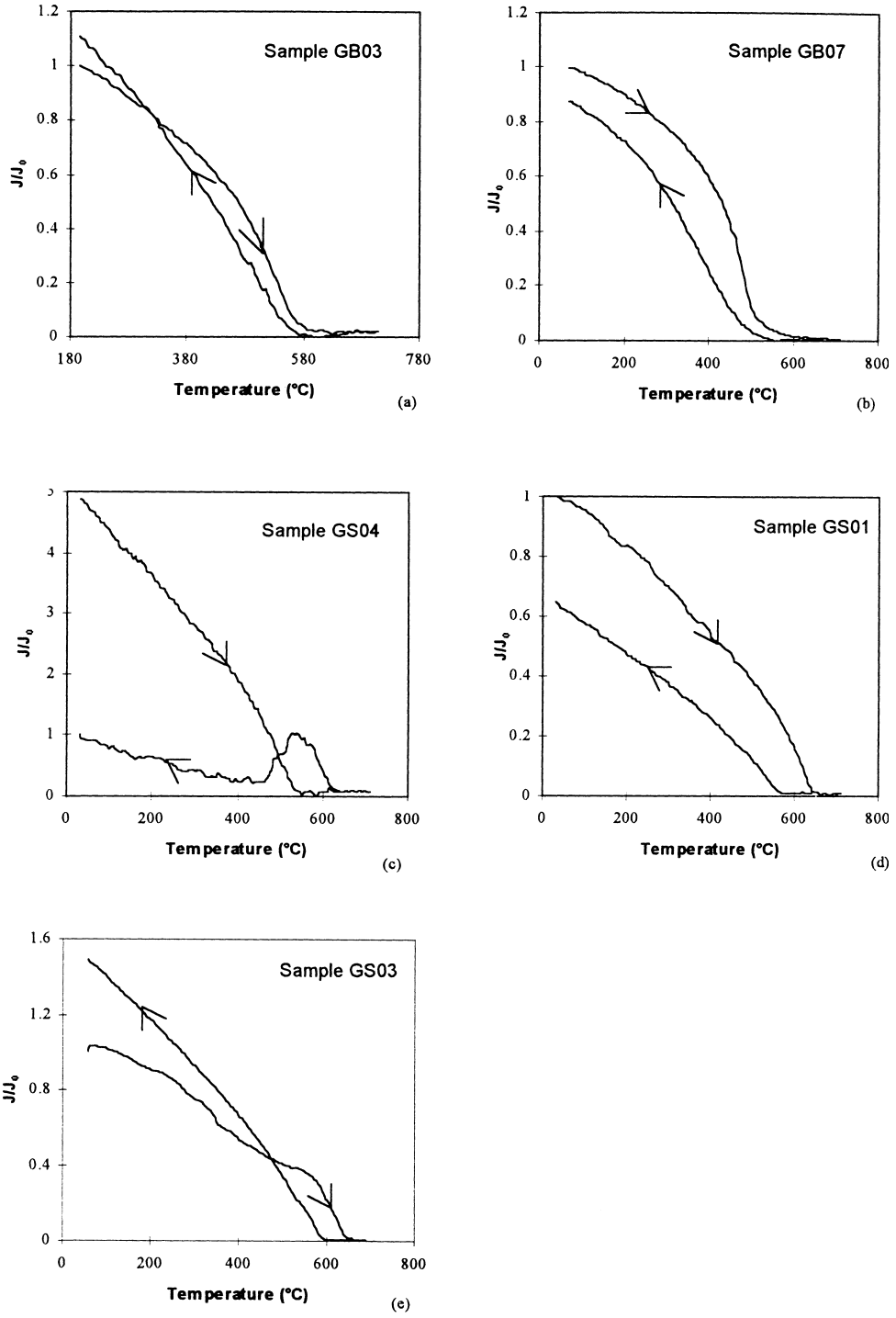


Figure 3. Representative thermomagnetic [$J_s(T)$] curves for the Jurassic intrusions (a),(b) Gingenbullen Dolerite; (c)–(e) Gibraltar Microsyenite.

curves (Fig. 6). All samples from sites GS1–3 showed a clear single component saturating at low applied fields (Fig. 6a) and with H_c and H_{cr} values expected for SD magnetite (Dunlop 1981; Thompson & Oldfield 1986). Curie temperatures for site GS1 (Fig. 3d) suggest the presence of CD magnetite, which exhibits the hysteresis properties of SD magnetite. The results for site GS4 (Fig. 6b) showed the presence of two magnetic mineral phases: a lower-coercivity magnetite phase and a higher-coercivity haematite phase.

3.5 Thermomagnetic (pTRM 'tail') criterion

As an additional test for the domain state of the remanence carriers in the samples, a new rock magnetic test (Shcherbakova *et al.* 1999) was conducted. A selection of fresh samples from both intrusions were heated above their Curie temperature (T_c) and cooled back to room temperature (T_r). During cooling from T_c to 300 °C, the samples were held in zero field; between 300 °C and T_r a 75 μT field was applied along the samples'

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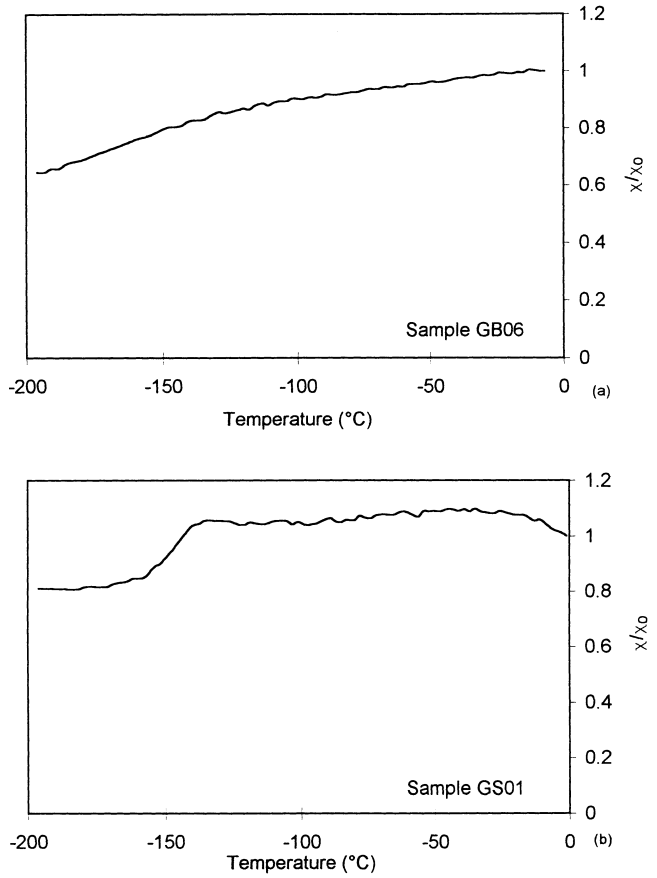


Figure 4. Representative low-temperature susceptibility (χ_{LT}: χ_{-196}/χ_{RT}) behaviour for samples from the intrusions. (a) Gingenbullen Dolerite; (b) Gibraltar Microsyenite.

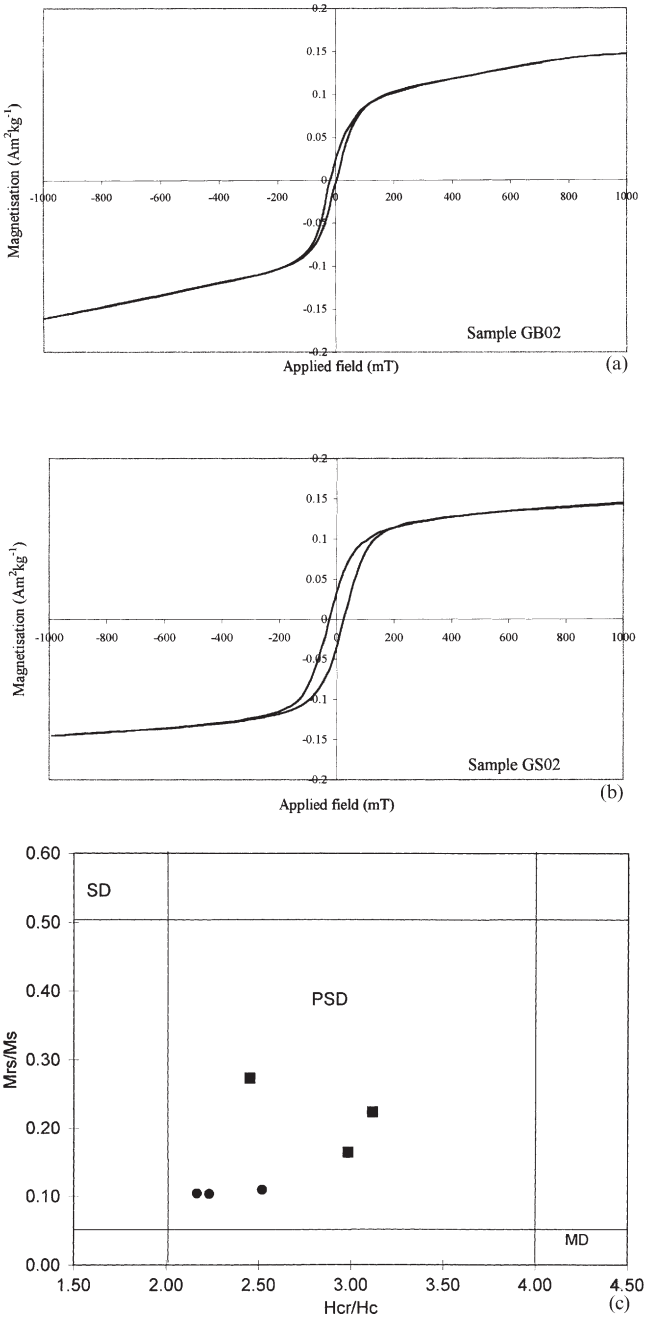


Figure 5. Representative hysteresis loops for the intrusions, (a) Gingenbullen Dolerite; (b) Gibraltar Microsyenite. (c) Day plot (Day *et al.* 1977) for representative samples from the Gingenbullen Dolerite (circles) and the Gibraltar Microsyenite (squares).

z-axes, imparting a $pTRM_{(300,\ Tr)}$ in this region. Once the magnetization of the samples was measured, they were again heated to 300 °C and cooled in zero field. The magnetization measured in the samples after this procedure was due to the tail of $pTRM_{(300,\ Tr)}$. The magnitude of this, relative to the total $pTRM_{(300,\ Tr)}$, is indicative of the domain state of the remanence carriers in the range the pTRM was imparted (Shcherbakova *et al.* 1999). This experiment was also repeated for the GB high blocking temperature (HBT) component using a $pTRM_{(500,\ Tr)}$.

The tail magnitudes for the GS $pTRM_{(300,\ Tr)}$ and the GB $pTRM_{(500,\ Tr)}$ (i.e. the ChRMs of both intrusions) were approximately 2 per cent of their original pTRM magnitudes (Table 1), implying that SD grains were the dominant remanence carriers in these temperature ranges. This indicates that the carriers of the ChRM components for the GB and the GS are, theoretically, ideal material for palaeointensity studies. The magnitude of the tail of $pTRM_{(300,\ Tr)}$ (i.e. the overprint) for the GB samples was approximately 10 per cent of $pTRM_{(300,\ Tr)}$ (Table 1), indicating that the average domain state of the remanence carriers in this range was PSD and hence suitable for Thellier-type palaeointensity studies.

3.6 Summary of rock magnetic results

The complete suite of rock magnetic experiments showed the GB dolerite to be theoretically more suitable for palaeointensity work due to the presence of a significant SD grain fraction for the ChRM and PSD for the overprint. Such grains are small enough to satisfy the necessary palaeointensity criterion of equality for blocking and unblocking temperatures (Bol'shakov & Shcherbakova 1979). The recognition that the magnetite phases present in most samples probably resulted from deuteric (high-temperature) oxidation and that any heating-induced alteration usually resulted in insignificant changes in J_s suggests that the majority of samples should have retained at least part of their primary TRM. The thermomagnetic curves and χ_{LT} results for the GS suggest, on face value, that this intrusion is less suitable for palaeointensity work. However,

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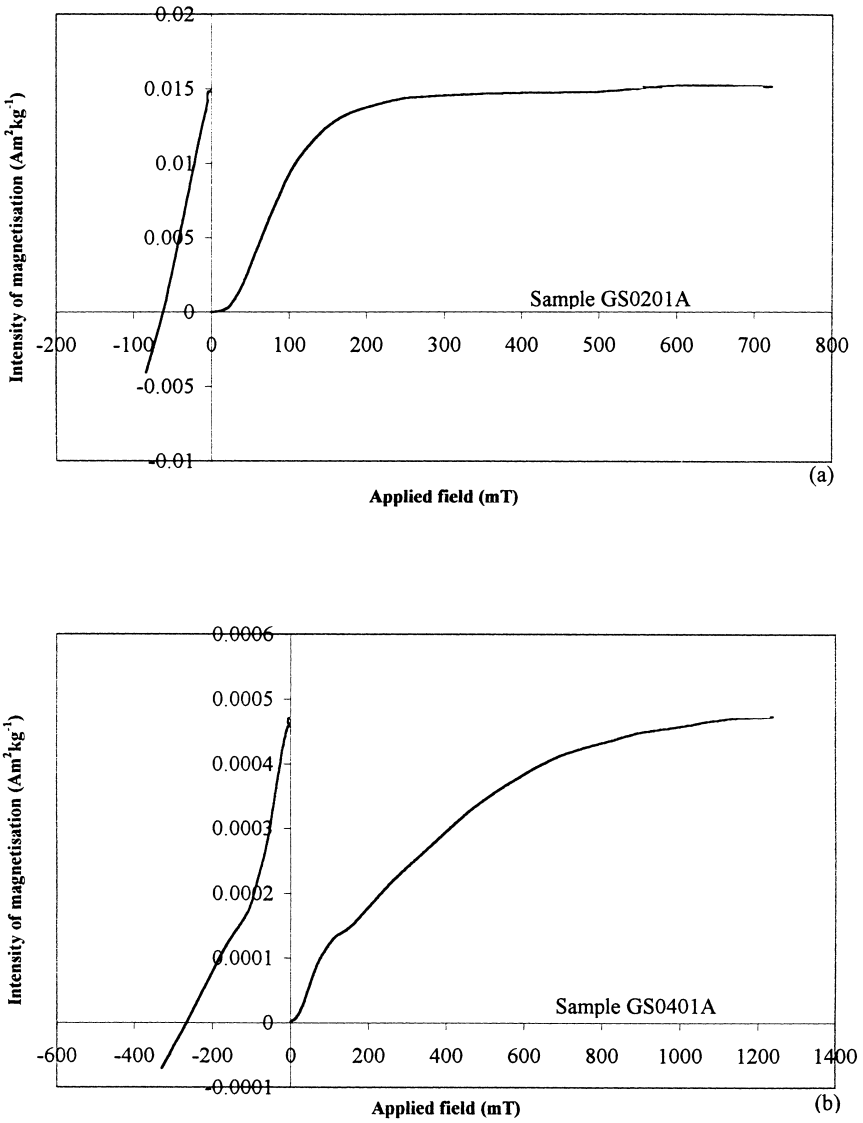


Figure 6. Partial remanent hysteresis curves for the Gibraltar microsyenite. (a) Site GS2; (b) Site GS4.

all samples lie in the PSD region of the Day plot and the new thermomagnetic criterion method shows the ChRM to be held by SD grains, indicating ideal material for palaeointensity work.

4 PALAEODIRECTIONAL RESULTS

Palaeodirectional work on the two intrusions was previously performed by Boesen *et al.* (1961), with a later study by Schmidt (1976). The results of the two studies are listed in Tables 2(a) and 3. Schmidt (1976) recognized the existence of two components of magnetization in the GB on AF demagnetization to 25 mT, the primary TRM being of reverse polarity and the overprint being a normally magnetized CRM or VRM. Schmidt (1976) attributed differences in the pole positions calculated by the two studies to the inadequate demagnetization of the secondary component in the earlier study. The remanence directions from the GS reported in the two studies are significantly different, with the Boesen *et al.* (1961) results (Table 2a) yielding a normal-polarity ChRM that is at variance with the reverse polarity reported by Schmidt (1976). Again, the difference is attributed to insufficient removal of secondary magnetizations in the earlier study.

The present study employed stepwise thermal demagnetization to determine the detailed remanence structure of samples from both intrusions. Samples were demagnetized (using a single-unit MMTD1 shielded demagnetizer) in steps of 50° from 100° to 500 °C, followed by steps at 530°, 560° and 580 °C. Where necessary, demagnetization was continued up to 680 °C to analyse any component residing in the haematite blocking temperature range. Remanence directions were measured using a Molspin Spinner Magnetometer. The results are illustrated in Figs 7 and 8.

Figs 7(a) and (b) illustrate a typical orthogonal vector (Zijderveld 1967) plot of thermal demagnetization for the GB. Two clear antiparallel components are observed: an HBT component defined between 450 and 580 °C and a low blocking temperature (LBT) component recognized below 400 °C. The distinctiveness of the two components strongly supports the

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Table 2(a). Palaeodirectional results from the Jurassic intrusives of southern NSW.

Intrusion	*N*	Δ*T*	*D* (°)	*I* (°)	α95°	*k*	Δ*T*	*D* (°)	*I* (°)	α95°	*k*
			LBT component					HBT component			
(a) Gingenbullen Dolerite											
GB1 (base of intrusion)	8		045	−68	15	18		107	+74	20	10
GB2 (margin)	6		026	−74	9	54		128	+78	7	87
GB3 (10 m above base)	4		344	−74	17	32		102	+85	17	54
GB4 (30 m above base)	5		257	−87	24	11		279	+83	17	22
GB5 (60 m above base)	6		349	−87	15	22		192	+81	17	17
GB6 (40 m above base)	7		024	−69	12	27		209	+58	15	18
GB7 (margin)	5		041	−82	18	18		183	+75	17	32
Mean (This study)	**41**	**150–400**	**021**	**−79**	**8**	**63**	**450–580**	**174**	**+81**	**11**	**33**
Mean (Schmidt 1976)	8							156	+79	central	
								142	+82	margin	
Mean (Boesen *et al.* (1961)								191	+80		
(b) Gibraltar Microsyenite											
GS1* (top of intrusion)	5		122	−2	14	4					
GS2 (~100 m above base)	4		196	+75	10	80					
GS3 (~90 m above base)	7		191	+70	2	757					
GS4 (~150 m above base)	2		076	+75	–	8					
Mean (This study)	**13**	**100–450**	**167**	**+79**	**24**	**26**					
Mean (Schmidt 1976)	8		142	+63							
Mean (Boesen *et al.* 1961)			359	−81							

N = Number of samples per unit; ΔT = dominant temperature range of remanence component; D = declination; I = inclination; α_{95} = semi-angle of 95 per cent confidence cone; k = precision parameter (Fisher statistics); * not used in calculation of mean direction.

Table 2(b). Mean directions and pole positions for Cretaceous overprint magnetizations from the Sydney and Bowen basins. References: 1 = Schmidt *et al.* (1995); 2 = Lackie & Schmidt (1996); 3 = Dunlop *et al.* (1997).

Location	Ref	*N*	Dec (°)	Inc (°)	α95
Central Queensland					
Hecate granite	1	20	20.8	−72.5	8.2
Blenheim granodiorites	1	10	315.1	−76.2	7.6
Harrow Creek Sill	1	9	21.1	−66.0	8.9
Bundarra granodiorite	1	4	235.7	−80.4	8.4
Daunia granodiorite	1	3	341.1	−74.0	10.1
Sydney Basin					
Norah Head	2	16	17.0	−80.3	3.8
Toowoon	2	36	6.4	−74.8	1.8
Forester North	2	36	17.9	−77.5	2.6
Otford	2	16	0.8	−79.1	5.8
Coalcliff	2	26	355.2	−74.4	3.7
Tahmoor	2	131	6.0	−76.4	1.6
Hornsby	2	70	5.5	−77.5	2.1
Dundas	2	13	349.7	−75.7	8.8
Marsden	2	36	43.5	−78.0	1.9
St Mary's	2	24	358.1	−81.8	2.6
Milton*	3	13	348.3	−79.0	6.5
Mean pole: Lat/Long = 56°S/147°E; dp/dm = 3.9/4.1					
		11	5.7	−78.1	2.2

presence of PSD/SD grains. The mean HBT direction for all sites (Table 2a) is $D/I = 174°/+81°$ (Fig. 8a), giving a virtual geomagnetic pole (VGP) position at Lat/Long = 52°S/153°E, which is close to those described by Schmidt (1976) (see Tables 2a and 3) and confirms this component as the

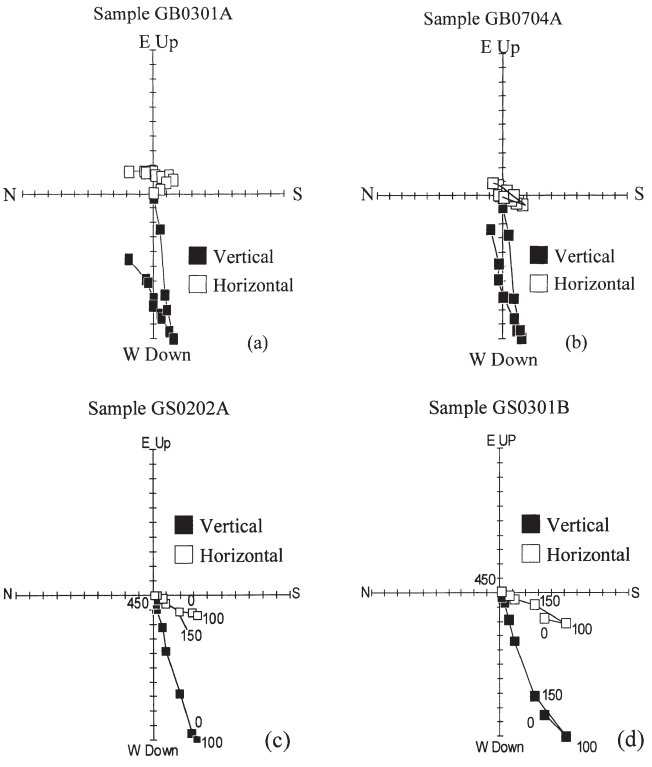


Figure 7. Representative palaeodirectional results for samples from (a),(b) the Gingenbullen Dolerite and (c),(d) the Gibraltar microsyenite. Orthogonal vector (Zijderveld) plots (OVPs) illustrate the full remanence component structure during thermal demagnetization. Open (closed) squares are projections onto the horizontal (vertical) plane.

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Table 3. VGP positions for the Jurassic intrusives of southern New South Wales.

Intrusion	Source	Pole Latitude (°S)	Pole Longitude (°E)	Pole errors *dp*, *dm* (°)
(a) Gingenbullen Dolerite				
ChRM (HBT component)	This study	52	153	17.3, 18.0
	Schmidt (1976) [central region]	53	164	
	Schmidt (1976) [margin]	46	165	
	Boesen *et al.* (1961)	53	144	
Overprint (LBT component)	This study	54	137	12.4, 12.9
(b) Gibraltar Microsyenite				
ChRM	This study	54	158	30.0, 31.0
	Schmidt (1976)	59	208	
	Boesen *et al.* (1961)	41	146	

primary TRM. The mean direction for the LBT component ($D/I = 021°/-79°$; Fig. 8b) is steeper than both the dipole and total present-field directions. The corresponding (VGP) position for this component is Lat/Long = 54°S/138°E, which is similar to Mid-Cretaceous pole positions recorded in rocks from the Sydney and Bowen Basins (Table 2b).

Tables 2(a) and 3 also document the results of palaeodirectional analyses for the GS. Samples from site GS1 displayed random ChRM directions, resulting in a poorly defined site mean. NRM intensities were very high, suggesting that lightning strikes have affected this site. Samples from site GS4 show the most noticeable weathering on core examination and, consequently, stable ChRM directions were isolated from only two cores from this site. Figs 7(c) and (d) show representative orthogonal plots from sites GS2 and GS3. A weak secondary component, with random directions, was typically present but removed by demagnetization to 100 °C or 150 °C. A convergent component remained until 450 °C, above which the signal became too noisy to yield a direction. The mean direction for the GS is $D/I = 167°/+79°$ (Fig. 8c), with a corresponding VGP position of Lat/Long = 54°S/158°E, the longitude of which differs from that of Schmidt (1976). Despite lying on the Jurassic segment of the Australian APWP (Dunlop *et al.* 1997), the limited number of sites used to calculate our pole precludes a meaningful comparison with other Jurassic results.

5 PALAEOINTENSITY STUDY

5.1 Experimental procedure

A total of 35 sister samples (20 from the GB and 15 from the GS) were selected for palaeointensity studies on the basis of having a well-defined remanence structure. The palaeointensity technique used was similar to that of Prévot *et al.* (1985), based on the Coe (1967) modification of the Thellier (Thellier & Thellier 1959) technique. The GB samples were heated in air using steps of 50 °C from 100 to a maximum of 600 °C, since magnetite was identified as the remanence carrier (Sections 3 and 4). For the GS samples heating steps were 40 °C from 100 to 460 °C, which was the temperature interval identified as being most likely to yield a meaningful palaeointensity result, based on the stability of remanence (Section 4).

After NRM measurement at each step, samples were reheated and cooled in air in the presence of a 25 μT applied field. PTRM checks were made at 100 °C intervals by demagnetization to a temperature 100 °C lower than that to which samples had just been remagnetized. Vector calculations were then made to check for laboratory-induced thermal alteration.

5.2 Data processing

The following data quality indicators were considered during the palaeointensity analyses:

(1) a minimum of four points, spanning at least 15 per cent of the total NRM, were used to define linear segments of the Arai (PNRM–PTRM) plots;

(2) the segment of the Arai plot that defines the palaeointensity should not show any systematic curvature, such that any subsegment of it could have a slope deviating from the overall segment slope by more than 20 per cent;

(3) segments affected by significant, unresolvable secondary components of remanence are ignored;

(4) PTRM checks within the defined linear segment should normally lie within 15 per cent of the original PTRM value.

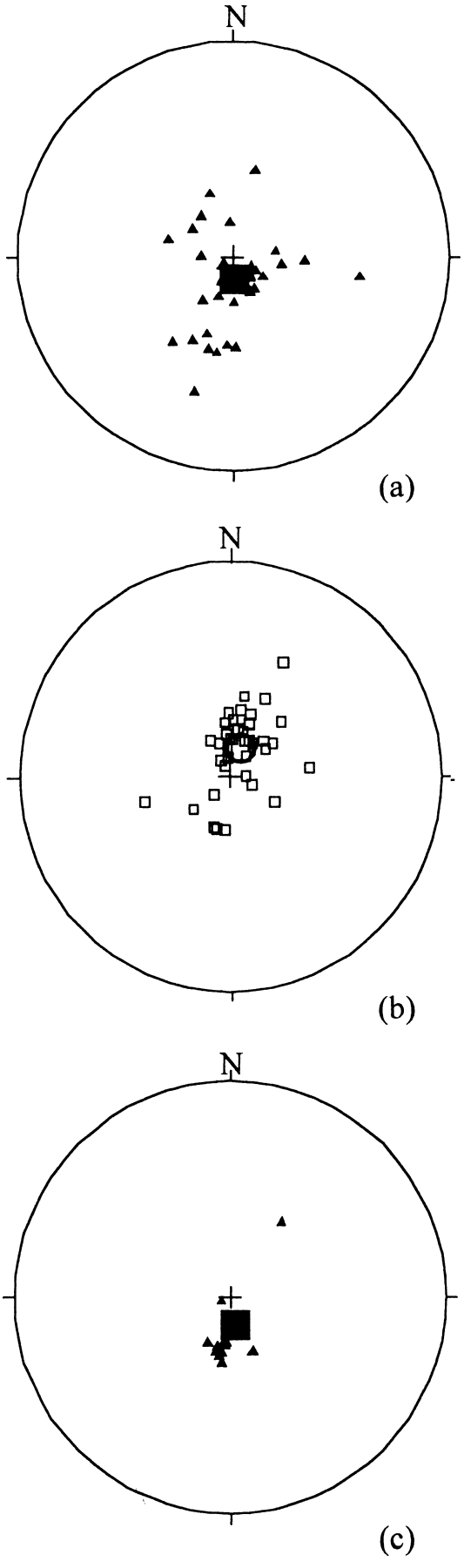
Samples were not rejected for failing point 4 *alone* if all other indicators were positive and the rock magnetic results indicated potential suitability.

5.3 Results

Table 4 shows palaeointensity results obtained from both the GB and GS, with associated quality statistics (Coe *et al.* 1978). Most of the GB samples showed behaviour similar to that illustrated in Fig. 9, with a positive slope in the low-temperature portion of the Arai plot, followed by a more gradual negative (expected) slope in the high-temperature portion. This is an artefact of the presence of two highly discrete, nearly antiparallel remanence components, where the lower-temperature component is removed below 400 °C. Therefore, the *scalar* magnitude of NRM decreases with increasing PTRM and can be used to give a palaeointensity result. Thus, two palaeointensity estimates can be obtained from the GB samples.

Several GB samples exhibited poor PTRM checks for the HBT component. However, these samples showed within-site consistency of palaeointensity values and had well-defined linear segments on Arai plots over the temperature intervals defined by the SD-bearing primary TRM. The Coe statistics (Coe *et al.* 1978; Table 4) also suggest results that are of reasonable to good quality. These factors combine to merit

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reportage of the results, the significance of which is discussed later. The results show low palaeointensity values and moderate q-values, with a weighted mean palaeointensity of 6.0 ± 1.4 μT, corresponding to a mean VDM of $(1.3 \pm 0.1) \times 10^{22}$ Am^2, suggesting that the geomagnetic field was approximately 16 per cent of its present value at 172 Ma.

Palaeointensity results from the low-temperature portions of Arai plots show a wide spread of values and generally have lower q-values than the HBT component. The weighted mean palaeointensity value (weighting as defined by Coe *et al.* 1978) is 39.8 ± 8.5 μT, with a corresponding range of VDM values from 3.5 to 6.5×10^{22} Am^2 (approximately 44–81 per cent of the present-day mean value).

Fig. 10 shows typical Arai plots from the GS. Alteration was witnessed at temperatures above 300 °C, as shown by a combination of non-linearity in the Arai plots and failure of PTRM checks. This alteration significantly reduced the portion of the blocking temperature spectrum available for recovery of a palaeointensity value. However, since any low-temperature overprint was normally removed by ~140 °C, it was possible to use up to six points (between 140 and 300 °C) for definition of the linear segment, a number sufficient to satisfy the Coe *et al.* (1978) criterion. Samples from sites GS1 and GS4 gave unstable Arai plots and therefore yielded no palaeointensity results. The palaeointensity results, with their associated statistical data, are listed in Table 4. The q-values range from 3.3 to 25.9, illustrating moderate to high quality, a fact reflected in the low uncertainty of the weighted mean palaeointensity of 26.6 ± 2.0 μT. This corresponds to a range of VDM values from 3.2 to 5.0×10^{22} Am^2, which is between approximately 40 and 62 per cent of the present-day mean VDM value.

The primary cooling histories of the two intrusions were estimated using the thermal model developed by Spear *et al.* (1991), which applies a finite difference algorithm to solve the heat transfer equation. Using this model, we calculate that the core and margins of the GB took ~13 000 years and ~5000 years, respectively, to reach ambient temperature (60 °C). Calculations for the larger GS suggest a corresponding cooling time of ~44 000 years for all parts of the intrusion. The natural cooling rates for the GB and GS are therefore 8 and 9 orders of magnitude higher than the laboratory cooling rate (60 min). Using the cooling-rate corrections developed by Fox & Aitken (1980) and Dodson & McClelland-Brown (1980), this difference would amount to an overestimate of palaeointensity of *up to* 56 per cent for the GB and 63 per cent for the GS. It is stressed that these calculations are only semi-quantitative and rely on assumptions of the physical parameters for the original melts and burial depths. Additionally, the theoretical cooling-rate corrections above are valid for SD grains, so the presence of PSD or MD grains would affect the correction. According to the rock magnetic results, the cooling correction presented here may only be valid for the ChRM (high-temperature component) of the GB. Therefore, no cooling-rate correction has been made for the GS, but it is noted that the palaeointensity values for this intrusion are also

Figure 8. Stereoplots of cleaned directions for (a) the Gingenbullen Dolerite overprint (low blocking temperature component), (b) the Gingenbullen Dolerite ChRM (high temperature component) and (c) the Gibraltar Microsyenite ChRM. Closed triangles indicate positive inclinations, open squares indicate negative inclinations.

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Table 4. Thellier palaeointensity results from the Jurassic intrusives of southern New South Wales.

Sample	ΔT	N	f	g	q	Ba (μT)	E	VDM
(a) Gingenbullen Dolerite (HBT component)								
GB102B	N/R							
GB105B	400–600	5	0.885	0.575	12.7	4.5*	1.1	0.64
GB201B	400–600	5	0.935	0.606	4.33	13.0	0.6	1.79
GB202B	400–600	5	0.942	0.538	4.65	11.5	0.7	1.58
GB203B	400–600	5	0.975	0.554	3.47	12.0	1.2	1.65
GB402B	400–550	5	0.838	0.715	7.43	5.8	0.8	0.77
GB404B	400–600	5	0.793	0.723	5.39	5.3*	1.9	0.70
GB405B	400–600	5	0.815	0.737	6.83	6.0*	2.6	0.79
GB501B	400–600	5	0.977	0.711	6.09	8.7	1.4	1.17
GB504B	400–600	5	0.992	0.698	6.29	5.9*	2.4	0.79
GB506B	400–600	5	0.939	0.668	4.38	4.9*	2.1	0.66
GB602B	400–600	5	0.878	0.607	2.83	7.0	2.5	1.23
GB603B	400–600	5	0.877	0.646	4.11	7.6	1.8	1.33
GB604B	400–550	4	0.844	0.584	4.40	9.2	2.1	1.61
GB605B	400–550	4	0.681	0.49	4.24	3.3	0.8	0.58
Mean						**6.0 ± 1.4**		**1.30 ± 0.11**
Cooling-corrected						**3.9**		**0.87**
(b) Gingenbullen Dolerite (LBT component)								
GB102B	250–450	4	0.521	0.483	1.57	28.0	1.2	4.30
GB105B	250–400	4	0.266	0.615	0.97	36.0	1.7	5.55
GB404B	200–350	4	0.208	0.524	0.68	27.0	4.0	3.51
GB501B	200–350	4	0.246	0.949	1.05	46.0	2.7	6.00
GB504B	200–350	4	0.400	0.652	2.10	50.0	2.5	6.50
GB605B	200–350	4	0.651	0.894	2.32	98.0*	42.0	14.91
GB704B	200–350	4	0.185	0.670	0.33	25.0	2.1	3.33
GB707B	200–350	4	0.251	0.583	0.94	35.0	2.4	4.66
Mean						**39.8 ± 8.5**		**4.84 ± 0.45**
(c) Gibraltar Microsyenite								
GS201D	180–300	4	0.440	0.640	3.75	35.0	1.8	4.96
GS202B	100–260	5	0.580	0.770	10.54	27.0*	1.6	3.83
GS203B	0–260	6	0.630	0.770	3.34	30.0*	2.0	4.25
GS302C	100–300	6	0.850	0.770	17.07	29.0	2.0	4.36
GS304D	140–300	5	0.660	0.720	7.00	27.0	3.8	4.06
GS305C	180–300	4	0.340	0.630	10.35	24.0	2.0	3.61
GS306C	140–300	5	0.630	0.700	4.27	28.0	2.1	4.21
GS307D	140–300	5	0.680	0.750	6.32	21.0	3.4	3.16
Mean						**26.6 ± 2.2**		**4.06 ± 0.3**

ΔT = temperature range used to calculate palaeointensity (linear Arai plot); N = number of data points defining the linear portion of the Arai plot; f,g,q = the NRM fraction, the gap factor and the quality factor, respectively, of Coe *et al.* (1978); Ba (μT) = palaeointensity value in microteslas; E = error in the palaeointensity estimate; VDM = virtual dipole moment corresponding to each site mean palaeointensity ($\times 10^{22}$ Am^2); LBT (HBT) = lower blocking temperature (higher blocking temperature); N/R = no result; * not used in calculation of mean value due to failure of PTRM checks in addition to unacceptably high errors in the mean palaeointensity value.

likely to be overestimates. The calculations are also only valid for the primary cooling of the intrusions and therefore no corrected palaeointensity value for the Cretaceous component in the GB can be made. However, by the same general principle, this value is also likely to be an overestimate of the true field strength. The corresponding cooling-corrected *minimum* mean palaeointensity and mean VDM values for the GB are listed in Table 4.

6 DISCUSSION

The GB and GS were selected as targets for this palaeointensity study because of their potential for recording palaeointensities for two distinct parts of the pre-Cenozoic Dipole Low. The ChRM for each intrusion is a primary TRM relating to initial cooling. The GB and GS are dated at 172 and 178 Ma, respectively, placing them within the 250–170 Ma time window for which data coverage is sparse. The secondary magnetization in the GB is now believed to be either a TCRM or, more probably, a TVRM derived from the 90 Ma thermal event involving uplift and supracrustal cooling related to the opening of the Tasman Sea. This event has been recognized from detailed palaeomagnetic studies (Schmidt *et al.* 1995; Lackie & Schmidt 1996; Dunlop *et al.* 1997), in addition to fission track dating (Morley *et al.* 1980; Marshallsea *et al.* 1985; Mallet *et al.* 1990), vitrinite reflectance (Middleton & Schmidt 1982), fluid inclusion and K/Ar dating (Bai *et al.* 1993). This 90 Ma event occurred in the main part of the Cretaceous Normal Superchron, so the palaeointensity results would be a valuable addition to the current global database.

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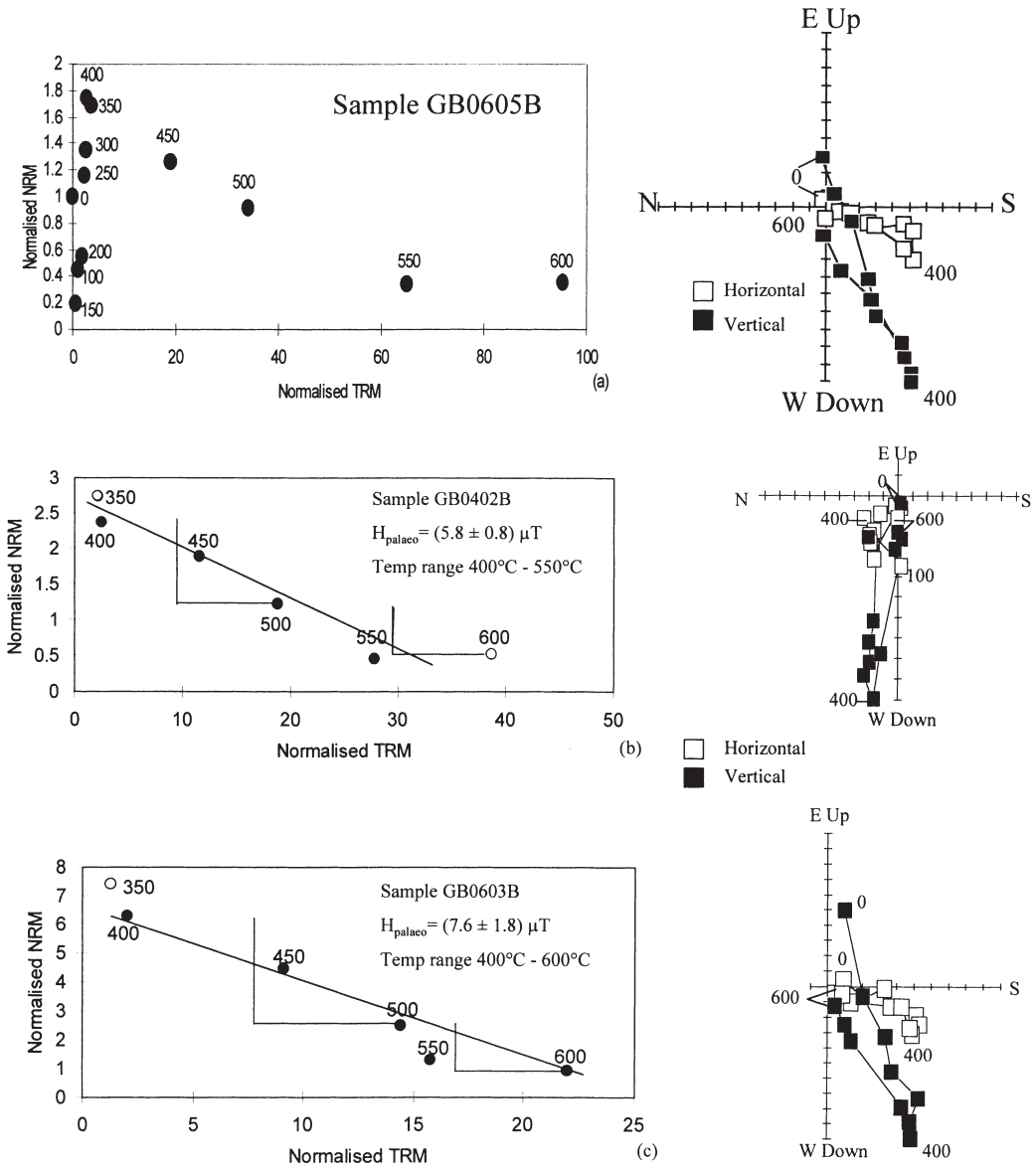


Figure 9. Representative Arai (NRM–TRM) plots from Thellier palaeointensity analyses of the Gingenbullen Dolerite. Filled (open) squares denote data points accepted (rejected) from the best-fit line used to define the palaeointensity value; Temp range is the temperature range over which the palaeointensity is calculated; PTRM checks are taken at elevated temperatures.

The remanence structure exhibited by the two intrusions is quite different: the GB records two distinct, antiparallel components with the LBT component (100–400 °C) being normally magnetized and the HBT (450–580 °C) recording a reverse polarity. The excellent separation of the two components is only possible if the remanence is held by SD/PSD grains. The remanence of the GS, on the other hand, is only stable up to a temperature of approximately 400–450 °C and records reverse polarity, with a mean direction very similar to that observed in the HBT component in the GB, although limited sampling means that we cannot be confident that the GS mean direction averages secular variation sufficiently. This steep-up direction gives a VGP that also correlates well with that observed by Schmidt (1976) for the GB and other Jurassic igneous rocks in Australia. Three alternative explanations can be offered for these observations: the respective ChRMs for the two bodies are initial cooling-related directions recorded in different parts of the blocking temperature spectrum; a geomagnetic field reversal occurred during the cooling of the GB, resulting in an HBT transitional direction and an LBT post-transitional normal direction; the ChRM recorded in the GS results from thermal overprinting, in the low to medium blocking temperature interval, related to the later intrusion of the GB. The last of these three alternative explanations is unlikely since the bodies cannot, on K/Ar dating evidence, be assigned to two separate magmatic events (McDougall & Wellman 1976; Embleton *et al.* 1985). Furthermore, the GB is likely to be too small to have a metamorphic aureole that affects the GS. Previous palaeomagnetic studies of the GB and from many other localities in the Sydney Basin (e.g. Schmidt 1976; Embleton *et al.* 1985; Schmidt *et al.* 1995; Lackie & Schmidt 1996; Dunlop *et al.* 1997) indicate the presence of

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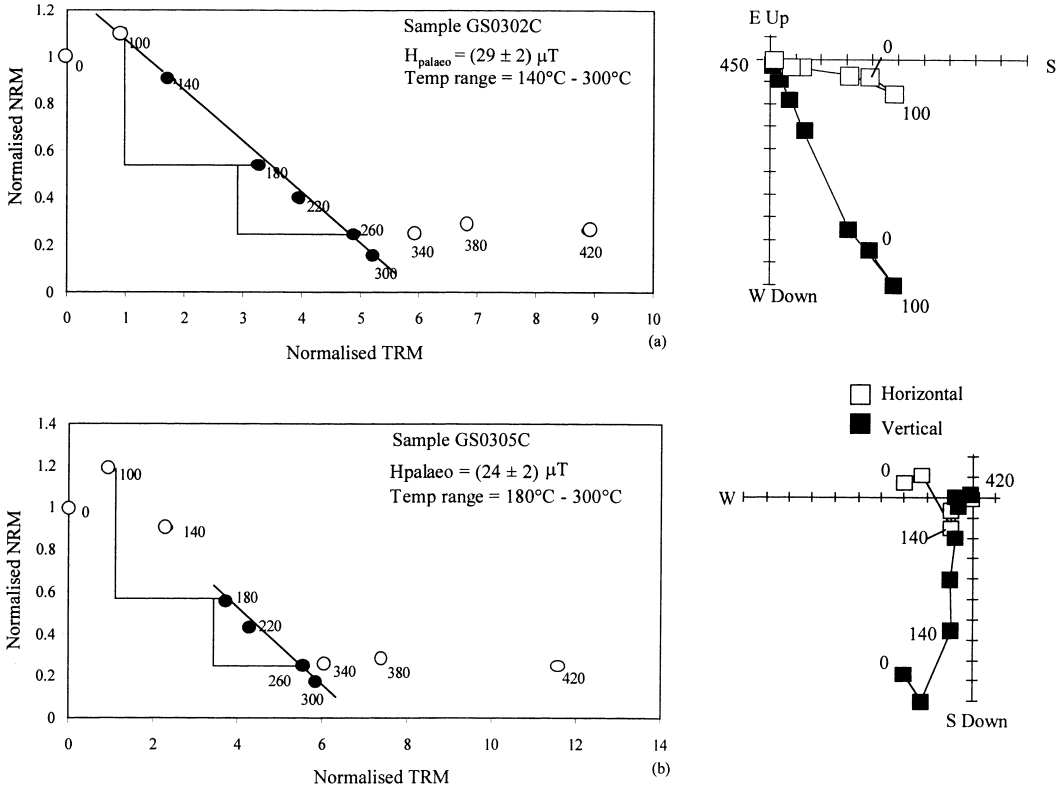


Figure 10. Representative Arai (NRM–TRM) plots from Thellier palaeointensity analyses of the Gibraltar Microsyenite. Filled (open) squares denote data points accepted (rejected) from the best-fit line used to define the palaeointensity value; Temp range is the temperature range over which the palaeointensity is calculated; PTRM checks are taken at elevated temperatures.

low-temperature overprints in rocks of different ages with different primary remanence directions. Therefore, the requirement for a transitional field is not supported by the published palaeomagnetic results. The first alternative is the more likely explanation, where the intrusions cooled over similar timescales and record the field characteristics during part of the second phase of igneous activity in the Sydney basin (Section 2). The NRM directions for the GS indicate that some later overprinting of the low blocking temperature region has occurred. This is removed by 100 °C and is likely to be related to Cretaceous or Tertiary rather than recent weathering, since directions do not correlate with the present or dipole fields (Schmidt 1976 and Section 4). There is no evidence to suggest that this low-temperature overprint in the GS has the same direction as the normal LBT component in the GB, suggesting that the magnetic mineralogy of this body was unaffected by the 90 Ma event, possibly due to deuteric oxidation during or immediately after initial cooling, resulting in the development of a resistance to remagnetization during reheating events.

The GB yields two mean palaeointensity results, one from the LBT component and one from the HBT component. The HBT result suggests that the field strength during initial cooling of the intrusion was very low, on average approximately 16 per cent of its present-day value. This is prior to the application of a cooling-rate correction, which further reduces the estimate to a minimum of approximately 11 per cent of the present value. The rock magnetic results (Section 3) clearly indicate the presence of SD magnetite as the remanence carrier in the GB ChRM, making it theoretically suitable for palaeointensity work. This is further supported by the clear separation of the overprint and ChRM components on vector plots (e.g. Fig. 7a). Furthermore, the palaeointensity results show quality values, as defined by Coe statistics (Coe *et al.* 1978), similar to or higher than those witnessed in other studies of ancient rocks (e.g. Perrin *et al.* 1991; Thomas *et al.* 1995, 1997, 1998) and are therefore considered meaningful. Notwithstanding, it is accepted that, due to the absence of consistently successful PTRM checks, the HBT palaeointensity results should be treated with caution. In keeping with evidence for SD magnetite remanence carriers from the thermomagnetic criterion test of Shcherbakova *et al.* (1999) and the location of samples on the Day plot, palaeointensity results from the GS are of a reasonable to high quality. However, the thermomagnetic (Curie) curves show that some alteration has taken place, possibly above the temperature region used for calculation of the palaeointensity results. This point confirms the need for caution when assuming that rock magnetic properties can be automatically used as a *solitary* rejection criterion during palaeointensity work. The relative VDM values for the GS samples range from approximately 40 to 62 per cent of the present-day value (with no cooling-rate correction), suggesting that, if the GS does indeed record a primary field, there is a large range of VDM values documented by these similarly aged intrusions. This is probably partly due to our inability to average out secular variation in the GS sufficiently, but on face value the results suggest that, over 6 Myr, the dipole field varies from at least 16 to 62 per cent of its present-day value. To put this variation into context, similar behaviour is witnessed throughout the Phanerozoic record and is therefore not unusual, illustrating the difficulty in pinning down long-term trends in

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the dipole moment. The Great Whin Sill of northern England records a dipole field strength some 80 per cent greater than the essentially contemporaneous Scottish Midland Valley Sill (Thomas *et al.* 1995). Perrin *et al.* (1991) noted that the dolerites on either side of the north Atlantic, dated at 200–190 Ma, record mean VDM values from 39 to 125 per cent of the present-day value, differences that cannot be attributed to secular variation. The mean VDM value for the last 5 Myr varies from 312 to 28 per cent of the present-day value (McFadden & McElhinny 1982), and a 70 per cent relative variation is witnessed for the past 12 000 years (McElhinny & Senanayake 1982). Fig. 11 shows the currently available palaeointensity data for the 250–150 Ma time window compiled from the palaeointensity database (Perrin 1998). Data from the present study provide palaeointensity estimates in a time window that previously contained limited Thellier data. These data support other results shown for the younger part of the 250–150 Ma window (Derder *et al.* 1989; Perrin *et al.* 1991) and confirm that the dipole moment was low during this part of the Mesozoic. These new results add weight to the possibility of a single dipole low during this time period, although data are urgently needed for the 210–250 Ma part of the window (Fig. 11 and gap 4, Fig. 1) to provide the key evidence.

The reliability of the palaeointensity results derived from the Cretaceous LBT component deserves further discussion. The *q*-values listed in Table 4 suggest that the results are not of the same quality as those derived from the HBT component but are, nonetheless, comparable with values from published studies of other Cretaceous rocks. Furthermore, the well-defined mean direction of magnetization recorded for the LBT component and the direct correlation of the temperature ranges used for definition of this component and calculation of the palaeointensity, in addition to favourable rock magnetic characteristics, also suggest that the field strength estimates are representative of the overprinting event. As previously discussed, there is a significant body of palaeomagnetic and other geological evidence to support the 90–100 Ma age of this overprint. Therefore, the data presented here are potentially important. The possibility of low-temperature alteration producing a CRM during such an event, leading to the production of maghemite, receives no support from the rock magnetic evidence (Section 3), which indicates PSD magnetite as the low-temperature remanence carrier in all GB sites. Therefore, available evidence suggests that the LBT may well be a TCRM or a TVRM, with maximum temperatures up to 400 °C. The resulting palaeointensity estimates can be considered seriously, whilst noting that a TCRM is likely to produce an overestimate of the field strength. Given this, the VDM values indicate that the field strength was also low (maximum range of approximately 44–81 per cent of the present value) during the approximately 90 Ma event, which falls within the CNS, values similar to those summarized in the global palaeointensity database for this period as a whole. Best estimates suggest that the major part of the event took place prior to 90 Ma (Morley *et al.* 1980; Bai *et al.* 1993), which places our LBT component VDM values within the sparsely covered 110–90 Ma part of the record. This very tentative result leads to the conclusion that the dipole field may also have been in a low-energy state during the superchron. This also appears to be the case for the older Permo-Carboniferous Superchron (Thomas *et al.* 1998). However, it must be stressed that the CNS result cannot be verified until we have estimates of dipole field strength that we can more positively identify as being derived from unequivocal TRMs within the 110–90 Ma window. Thellier studies on several igneous intrusions of this age from the Bowen Basin in Queensland and the northern part of the Sydney Basin that are known to hold a true primary TRM (Lackie & Schmidt 1996) are currently in progress. The growing global palaeointensity database appears to suggest that there is no direct link between geomagnetic field strength and reversal frequency. Results from these new studies will add to our understanding of field behaviour during the bulk of the CNS and allow us to contribute significantly to the field strength–reversal frequency debate.

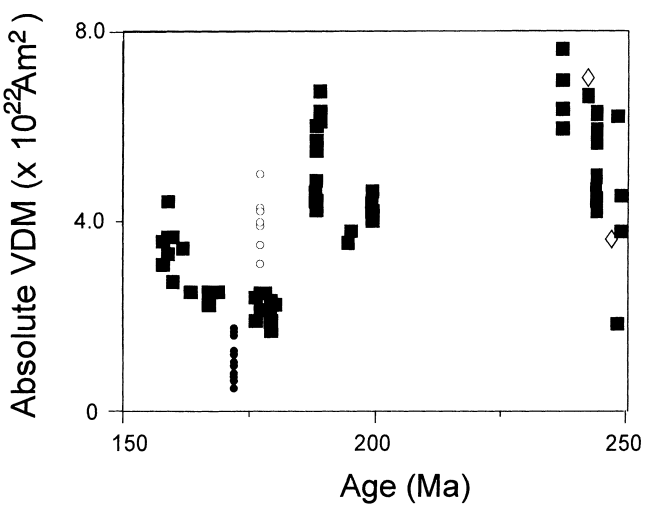


Figure 11. VDM results for the 250–150 Ma time window. Data points: Perrin (1998) (closed squares), the Kiaman project (Thomas *et al.* 1998, 1997, 1995; Harcombe-Smee *et al.* 1996) (open diamonds) and this study (closed and open circles).

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